

Aaditya Bhatnagar

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Education

- **Birla Institute of Technology and Science Pilani, Dubai Campus, Dubai, UAE** **CGPA: 9.25/10.00**
Bachelor of Engineering (B.E.) in Computer Science *Expected Sep 2026*
 - **Computer Science:** Generative AI, Natural Language Processing, Deep Learning, Data Mining, Software Architecture, Data Structures & Algorithms, Database Systems, Operating Systems
 - **Mathematics and Statistics:** Numerical Analysis, Linear Algebra, Probability & Statistics, Calculus
 - **Finance and Economics:** Derivatives & Risk Management, Security Analysis & Portfolio Management

Professional Experience

- **Merck Sharp & Dohme (MSD) GCC** **Dubai, UAE**
Data Science Intern *Jul 2025 – Jan 2026*
 - Built and maintained production ML pipelines for medical sales forecasting in Dataiku DSS, generating validated outputs with different algorithms like XGBoost, LightGBM, and Random Forest for the 2026–27 regional planning cycle.
 - Developed a Dataiku API-client interface for authenticated dataset modification, model scoring, parameter updates, and business scenario testing for custom values based on stakeholder requirements.
 - Automated forecasting dataset creation and validation using Python and Dataiku DSS, reducing manual preparation for production planning workflows, reducing time cost from days to minutes.
 - Created exploratory Dataiku visualizations to communicate medical sales forecasts and scenario outputs.
 - Fixed automated production scenario-pipeline failures caused by deprecated functions, preserving the data quality of the forecasting dashboards.
 - **Dataiku Certified:** Core Designer, Advanced Designer, ML Practitioner, MLOps Practitioner

Research Projects

- **Machine Unlearning in Vision-Language Models** **Feb 2026 – Present**
CLIP, Unlearning, LoRA, SLURM
 - Built a machine unlearning pipeline for frozen CLIP vision-language models tested using CIFAR-10 and CIFAR-100 retain/forget splits, storing adapter-only checkpoints.
 - Implemented parameter-efficient unlearning with CLIP ViT-B/16 vision LoRA and H-TGSD for CIFAR-100 flower-superclass forgetting and selective rose forgetting.
 - Compared H-TGSD against GA+KL, retain-only, counterfactual/entropy rebind baselines, and no-sibling ablations to study utility, forgetting, and sibling-class preservation.
 - Evaluated results with retain accuracy, target forget accuracy, sibling accuracy, membership-inference attacks, semantic-subspace metrics, experiment tracking, and tradeoff plots.
 - Reported preliminary selective-rose results with 0.6% forget-train accuracy, 11.0% rose-test accuracy, 91.75% sibling flower accuracy, and 86.32% retain accuracy.
- **Finetuning Defence Strategies for Adversarial Datasets** **Feb 2025 – Jun 2025**
DistilBERT, InfoBERT, RanMASK, ANLI, BReLU, RGeLU, t-Sigmoid
 - Analyzed adversarial NLP threats across character, word, sentence, and multi-level perturbations, using ANLI dataset to benchmark robustness under adversarial examples.
 - Implemented & compared bounded intermediate activations across DistilBERT, InfoBERT, and RanMASK variants.
 - Introduced **RGeLU**, which improved InfoBERT ANLI accuracy from 32.2% to 48.1%, exceeding the 46.9% DistilBERT-ANLI baseline.

Technical Projects

- **GroundDesk: Evidence-Grounded Support Analytics Copilot** ground-desk.onrender.com
Python, FastAPI, Gemini API, Qdrant, Supabase PostgreSQL
 - Built a B2B support analytics product for internal teams with Markdown, TXT, PDF, and URL ingestion; hybrid retrieval; cited Gemini responses; PostgreSQL answer traces; and safe escalation for insufficient evidence.
 - Developed labelled evaluation pipelines for retrieval and grounded response behavior, measuring citation grounding, expected answer-term coverage, escalation decisions, and unsupported-query handling.
 - Achieved 100% correct top citation for answerable cases, 93.8% expected answer-term coverage, and 100% unsupported/ambiguous escalation accuracy on a product-specific benchmark.
- **MLP Trainer + Agentic LLM Tutor**
Rust, Axum, Tokio, React, TypeScript, FastAPI, Groq API, MCP, Langfuse, OpenTelemetry, Docker
 - Built a 2-4-1 MLP visualizer with core logic in Rust, implementing forward/backpropagation, MSE loss, SGD, seed value for deterministic initialization, and support for XOR, half-moons, and spiral datasets.

- Exposed training and inference through an Axum/Tokio control plane and React interface for live weights, activations, loss curves, and predictions. Added an agentic FastAPI layer using Groq tool calling, bounded multi-turn memory, SSE streaming, MCP integration, Langfuse/OpenTelemetry, and containerized Rust, Python, and React services.
- **DeepLense: Gravitational Lensing Image Classification (ML4Sci GSOC Test)**
PyTorch, scikit-learn, ESCNN, CNNs, ROC-AUC
 - Trained a C8-equivariant CNN for 3-class gravitational lens classification to predict dark matter substructure, achieving 0.9862 mean validation AUC.
 - Trained a VGG-style CNN for imbalanced binary lens detection, achieving 0.9912 ROC-AUC and identifying 191/195 test lenses.
- **QNT: Regime-Aware Quantitative Research Platform**
Python, Pandas, scikit-learn, SQLite, Parquet, Streamlit
 - Built a real-data quant research CLI that loads Yahoo Finance prices and optional FRED macro data, creates point-in-time features, and generates momentum and walk-forward Ridge signals.
 - Ran long-only portfolio backtests with turnover costs and risk statistics including Sharpe, drawdown, VaR/CVaR, turnover, and NAV.
 - Exported Parquet artifacts, SQLite analytics views, HTML reports, deterministic agent reports, optional MLflow logs, and a Streamlit interface for backtest inspection.
- **NRG: Spanish Power Price Forecasting and Reporting Platform**
Python, SQL, PostgreSQL, scikit-learn, MLflow, FastAPI, React, TypeScript
 - Built a Spanish power-price research platform that loads raw CSV data into PostgreSQL, validates it with SQL quality checks, exports Parquet marts, and trains MLflow-tracked time-series models with walk-forward evaluation.
 - Compared temporal baselines and ML models for next-day average price forecasting, with Ridge reaching 2.23 EUR/MWh test MAE and 2.24 EUR/MWh walk-forward MAE.
 - Exposed forecasts, validation-selected backtests, model runs, data status, and AI assistant queries through FastAPI endpoints and a React/TypeScript dashboard.

Technical Skills

- **Languages:** Python, SQL, C++, Rust, TypeScript, R
- **Machine Learning:** NumPy, Pandas, PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, CLIP, LoRA, ESCNN
- **AI & Agents:** RAG, MCP, Gemini API, Groq API, tool calling, SSE streaming, Langfuse, OpenTelemetry
- **Software & Data:** FastAPI, Axum, REST APIs, Docker, Git, CI/CD, PostgreSQL, SQLite, SQLAlchemy, Parquet, Dataiku DSS, Qdrant, Supabase, MLflow, SLURM
- **Analytics:** Matplotlib, Seaborn, Tableau, Streamlit, Excel, yfinance, FRED

Achievements

- **1st Place, BITS Tech Fest Hackathon** – Built RescuMe, an emergency response MVP using geolocation mapping for disaster coordination between rescue teams and victims.